

William PENSEC

Associate Professor

 October 26, 1996

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 Driving licence

 French

Social Network

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 WilliamPsc

Languages

 French

 English

 Italian

Computer Science

 Software development

C, Python, C++, Java, JavaScript, Android, Bash, PHP, Ada, Visual Basic

 Hardware development

System Verilog, Assembly, RISC-V

 Data management

MySQL, JSON, XML

 Operating systems

Linux, Windows

 Microcontrollers

Arduino, Raspberry Pi, STM32, FPGA

 Miscellaneous

Latex, TikZ, Git, Keras

Interests

 Sport

Swimming, biking, hiking, nautic activities

 Hobbies

Travel, video games, movies, TV series, music, reading

About me

I am currently working as associate professor in Computer Engineering at LIRMM in University of Montpellier (France). I am specialised in the field of computer architecture, hardware security and fault injection attacks in embedded systems (specially on RISC-V).

Working Experience

2025 -	Associate Professor	LIRMM - Univ. Montpellier
	Member of ADAC team - Microelectronic's department	
2024 (1 an)	Postdoctorate fellow	Laboratory Hubert Curien - Saint-Etienne
	Security Evaluation of Neural Network FPGA implementations	
2023 (5 mois)	International mobility	ALaRI - Lugano, Switzerland
	Study and development of countermeasures to protect a system against fault injection attacks (SystemVerilog).	
2021 (5 months)	Final year internship (M2)	Lab-STICC
	Cooperation of drones in a heterogeneous system (C++, Linux, Java, TCP/IP, Raspberry Pi, embedded CNN)	

Education

2021 – 2024	PhD Thesis (Lab-STICC)	University of South Brittany - Lorient
	Enhanced Processor Defence Against Physical and Software Threats by Securing DIFT Against Fault Injection Attacks - European Label	
2019 – 2021	MSc in Software for Embedded Systems	University of Western Brittany - Brest
2015 – 2019	BSc in Computer Science: Fundamentals and Applications	University of Western Brittany - Brest
2014 – 2015	First Year of Medicine	University of Western Brittany - Brest
2014	Baccalaureate S-SVT, speciality ISN	High School of Cornouaille - Quimper

Papers

2024	Scripting the Unpredictable: Automate Fault Injection in CABA Simulation for Vulnerability Assessment – W. Pensec, V. Lapôtre and G. Gogniat – DSD
2024	Exploring Fault Injection Attacks on CVA6 PMP Configuration Flow – K. Quénehervé, W. Pensec, P. Tanguy, R. Dafali et V. Lapôtre – DSD
2024	Defending the Citadel: Fault Injection Attacks against Dynamic Information Flow Tracking and Related Countermeasures – W. Pensec, F. Regazzoni, V. Lapôtre and G. Gogniat – ISVLSI
2023	Another Break in the Wall: Harnessing Fault Injection Attacks to Penetrate Software Fortresses – W. Pensec, V. Lapôtre and G. Gogniat – Sensors S&P
2022	Smart Anomaly Detection and Monitoring of Industry 4.0 by Drones – W. Pensec, D. Espès, C. Dezan – ICUAS

Supervision

2025 –	Co-supervision of a PhD student: Ali Ait Hassou – ANR Project SCREAM	University of Montpellier
2026	Supervision of a second-year engineering student	University of Montpellier
	Implementation of an automatic mechanism to classify data according to the context	
2022	Supervision of a M2 student in project	University of South Brittany
	Implementation of a RISC-V processor on an FPGA target	